

FreshTrack

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Project Objectives

- Problem Statement
- Project Overview – Introduction
- End Users
- Wow Factor in Project
- Modelling/Block Diagram/Flow of Project
- Result/outcomes
- Conclusion
- Future Perspective



Problem Statement

Household food waste is one of the most overlooked contributors to environmental and economic loss worldwide — nearly a third of all food produced globally is wasted, much of it inside individual homes rather than farms or supply chains.

Most households have no structured way to track groceries. Without visibility into what's already in the fridge, items get pushed to the back, expiry dates go unnoticed, and usable food is thrown away — not from carelessness, but a lack of awareness and reminders. This is worse in shared kitchens like hostels, mess canteens, and small restaurants, where manual tracking is harder and commercial inventory systems are too costly.

The impact goes beyond the kitchen: every wasted item wastes the water, energy, land, and labor spent producing it, and its decomposition in landfills releases methane, a potent greenhouse gas — while households quietly lose a share of their monthly grocery budget to food never consumed.

Yet no simple, low-cost tool exists to help everyday households track what they own, know how long it will last, and get nudged to use it in time — existing options demand expensive smart-kitchen hardware or complex commercial software.

1/3

of food produced worldwide is wasted

\$1,000+

lost per household every year to food waste

8-10%

of global greenhouse emissions linked to food waste

Project Overview - Introduction

FreshTrack is an AI-powered kitchen assistant that turns food-waste prevention into a habit instead of a guessing game. Users photograph groceries or a receipt when they shop — the system identifies items, estimates how long each will stay fresh, and keeps a running “use-it-soon” list so nothing is forgotten at the back of the fridge.

- ☐ Snap a photo of groceries or a receipt — no manual typing
- ☐ AI estimates shelf life per food category
- ☐ Daily/weekly reminders before items spoil
- ☐ Recipe ideas built around what's about to expire

Use-It-Soon

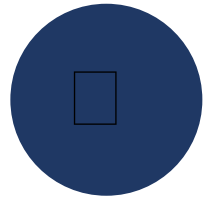
Milk 2 days left

Spinach 1 day left

Tomatoes 4 days left

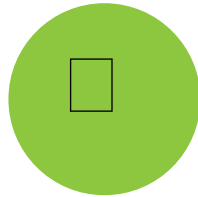
Bread 3 days left

End Users



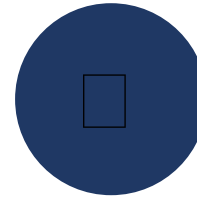
Households

Families wanting to cut grocery bills
and waste less week to week



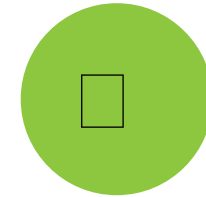
Hostels & Mess Canteens

Shared kitchens tracking bulk
perishables for large groups



Small Restaurants & Cafés

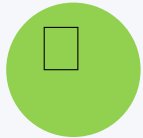
Managing perishable inventory
without expensive POS systems



Sustainability Communities

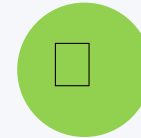
Apartments and eco-groups running
waste-reduction drives

Wow Factors in Project



Zero manual entry

One photo of groceries or a receipt logs the whole haul — OCR + image recognition do the rest.



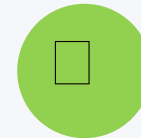
Predictive, not reactive

Shelf-life estimates warn users days before spoilage, not after it's already wasted.



Turns waste into meals

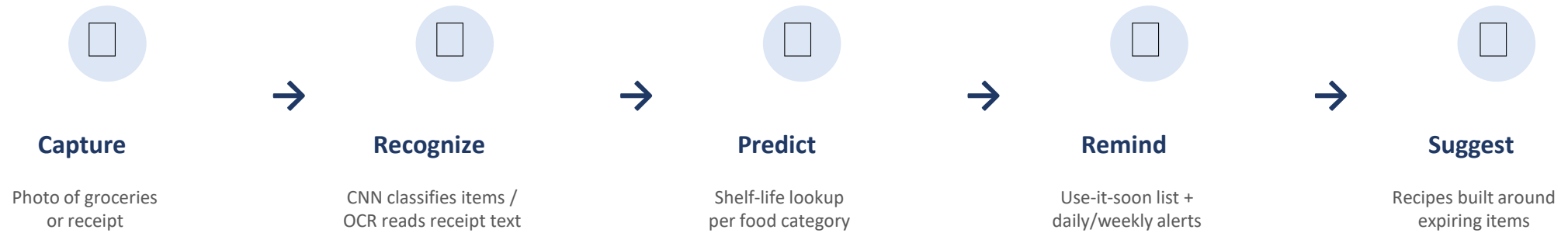
Expiring items automatically surface as recipe suggestions instead of trash.



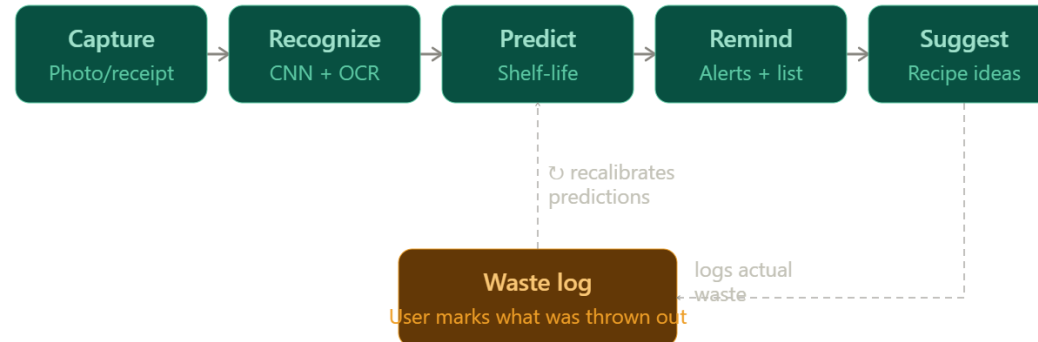
Learns from real behavior

The waste log lets users mark what was actually thrown out, sharpening future predictions.

Modelling/Block Diagram/Flow of Project



Feedback loop: users mark what was actually thrown away in the Waste Log — the shelf-life model recalibrates over time.



Technology Stack

Software & AI

- Python — core language
- Scikit-learn(Random forest + K-means) –shelf-life predication, waste-risk classification, and risk clustering
- Pandas/ NumPy – data handling
- Matplotlib – Visualizations
- CNN (Teachable Machine stand-in) – item recognition from photos, swapable for a real trained model
- OCR— reads item names from grocery receipts

Hardware

- Smartphone or laptop camera only
- No sensors or microcontrollers required
- Runs on everyday consumer devices

Result/Outcomes

```
1. Add grocery item (ML predicts shelf life + waste risk + cluster)
2. View 'Use-It-Soon' reminder list (ML-prioritized)
3. Get recipe suggestions for a category
4. Mark an item as used
5. Log an item as wasted (feeds back into ML predictions)
6. View waste summary
7. View full inventory
8. View ML model report (accuracy, R^2, clusters)
9. Upload/Parse grocery bill (auto-import all items)
10. Exit
=====
Choose an option (1-10): 9
Enter path to grocery bill text file: "C:\Users\keval & vasu\Downloads\Grocery_Bill.txt"
Error: Bill file "C:\Users\keval & vasu\Downloads\Grocery_Bill.txt" not found.

===== FreshTrack - AI/ML Edition =====
1. Add grocery item (ML predicts shelf life + waste risk + cluster)
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9. Upload/Parse grocery bill (auto-import all items)
10. Exit
=====
Choose an option (1-10): |
```

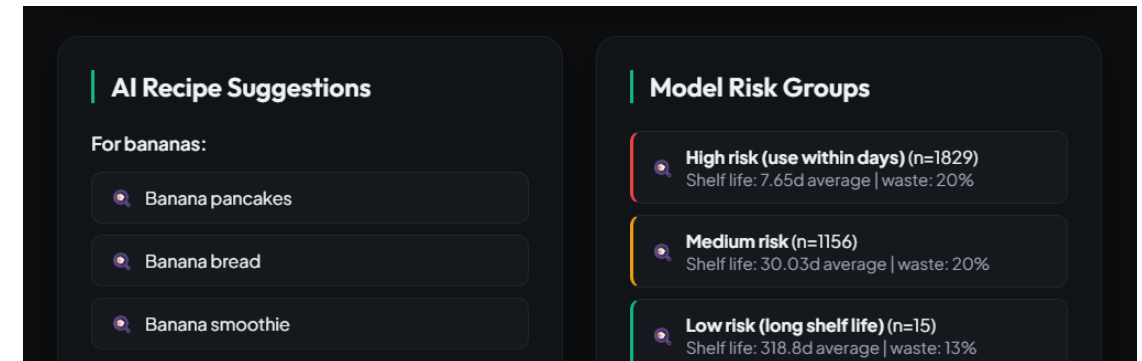
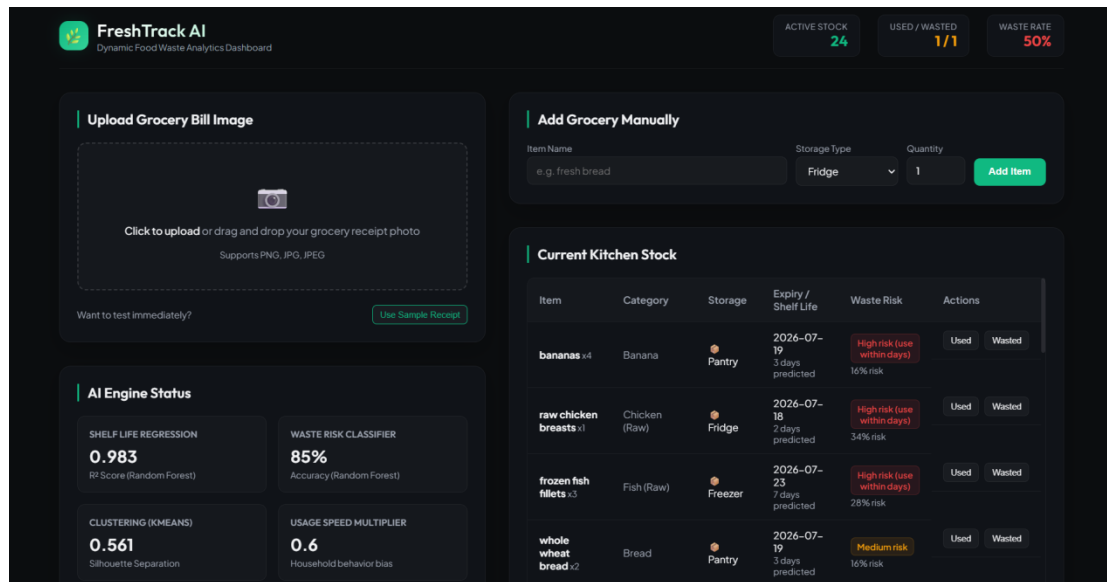
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\keval & vasu\Downloads\ml\freshtrack_ml\freshtrack_ml> python freshtrack_ml.py

===== FreshTrack - AI/ML Edition =====
1. Add grocery item (ML predicts shelf life + waste risk + cluster)
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10. Exit
=====
Choose an option (1-10): |
```

Result/Outcomes



Result/Outcomes

```
9. Upload/Parse grocery bill (auto-import all items)
10. Exit
=====
Choose an option (1-10): 3
Pick a category:
1. leafy greens
2. berries
3. banana
4. apple
5. tomato
6. citrus
7. milk
8. yogurt
9. cheese
10. eggs
11. bread
12. chicken (raw)
13. fish (raw)
14. cooked leftovers
15. rice (cooked)
16. root vegetables
17. packaged snacks
Enter number: 9
Recipe ideas for 'cheese':
- Grilled cheese sandwich
- Cheesy omelette
- Quesadilla
```

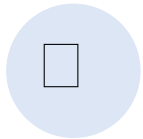
```
Choose an option (1-10): 7
Current inventory:
- organic spinach (leafy greens, fridge) | use by 2026-07-21 | waste risk 24% | High risk (use within days)
- fresh whole milk (milk, fridge) | use by 2026-07-23 | waste risk 23% | High risk (use within days)
- bananas (banana, pantry) | use by 2026-07-19 | waste risk 16% | High risk (use within days)
- raw chicken breasts (chicken (raw), fridge) | use by 2026-07-18 | waste risk 34% | High risk (use within days)
- frozen fish fillets (fish (raw), freezer) | use by 2026-07-23 | waste risk 28% | High risk (use within days)
- whole wheat bread (bread, pantry) | use by 2026-07-19 | waste risk 16% | Medium risk
- sweet apples (apple, pantry) | use by 2026-07-27 | waste risk 28% | Medium risk
- potato (root vegetables, pantry) | use by 2026-07-28 | waste risk 28% | Medium risk
- milk (milk, freezer) | use by 2026-08-09 | waste risk 29% | High risk (use within days)

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```

Conclusion

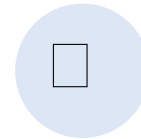
- ✓ Household food waste is a solvable, everyday problem — not just an industrial-scale one.
- ✓ FreshTrack lowers the effort of tracking groceries to a single photo, making waste-prevention a habit rather than a chore.
- ✓ A lightweight AI stack (image recognition + OCR + a shelf-life lookup) is enough to deliver real value without heavy infrastructure.
- ✓ Early pilot targets point to meaningful cuts in waste, household spending, and avoidable emissions.
- ✓ SDG 12- Consumption and production, this model tackles consumer-level food waste.

Future Perspective



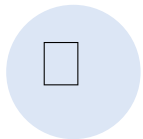
Smart fridge integration

Sync directly with smart-fridge sensors for automatic inventory updates



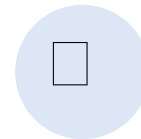
Community food-sharing

Let neighbors claim surplus food nearing expiry instead of discarding it



Retailer partnerships

Pull structured expiry data straight from grocery e-receipts and loyalty apps



Regional shelf-life models

Localize predictions for climate, diet, and produce variety by region

Thank You

Video (google drive link):

https://drive.google.com/drive/folders/15SDlwUUQadIT8M_81i2wZHFfa7OJA27E2

GitHub link:

https://github.com/jayvina/Edunet_CapstoneProject_FreshTrack